INTERNATIONAL MONETARY FUND

ISLAMIC REPUBLIC OF AFGHANISTAN

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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This debt sustainability analysis (DSA) assesses Afghanistan's external and overall risk of debt distress as high.[1] Afghanistan's debt sustainability depends on donor grants inflows (currently around 38 percent of GDP) to finance underlying fiscal and external deficits. Both external and public debt will remain low and mechanical signals over the 10-year horizon suggest moderate risk of debt distress. However, as grants are expected to decline over the medium and long term and be replaced by debt financing, the mechanical risk ratings based on extended 20-year periods are used. One of these—the ratio of present value of debt to exports—breaches the threshold under the baseline scenario, suggesting the high-risk rating. Political uncertainty, threat of violence, deeper than expected drop in aid, and weather shocks all provide significant downside risks. The authorities should continue improving revenue mobilization and strengthen debt management capacity, including effective monitoring of state-owned corporations (SOC) and public private partnerships (PPPs). In addition, Afghanistan should continue improving its debt carrying capacity through diversification of exports and strengthening debt management, including through local debt market development.

[1] This joint World Bank/IMF Debt Sustainability Analysis (DSA) has been prepared in the context of the 2020 request for emergency financing from the Fund, RCF. The macro framework underlying this DSA is the same as that included in the staff report of the 2020 RCF request which reflects recent global and domestic developments. The current macroeconomic framework reflects currently available information. However, updates with respect to economic impact and policy response to the COVID-19 crisis are rapidly evolving and risks are heavily tilted to the downside.

Islamic Republic of Afghanistan Joint Bank-Fund Debt Sustainability Analysis[1]	
Risk of external debt distress:	High[2]
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	Yes: The projection period of informing mechanical risk signals is extended to 20 years.
Macroeconomic projections	In the near and medium term, real GDP growth will be slower compared to the last DSA due to COVID-19 impact (average of 3.0 percent in 2020–25 compared to 3.7 in 2019–24 from the last DSA), and primary deficits will be larger (average of 1.1 percent compared to 0 in the above-mentioned periods). Revenue loss is the key driver of the higher deficits. In the long run, there will be slower real GDP growth (4.0 percent compared to 4.5 percent); lower grants (4.2 percent compared to 6.7 percent of GDP) and expenditure (23.6 percent compared to 26 percent of GDP) than in the previous DSA.[3] Debt relief under the Catastrophe Containment and Relief Trust (CCRT) is reflected in the exceptional financing line in Table 1 and in the debt relief (HIPC and other) line in Table 2.
Financing strategy	In the near term, larger deficits will be financed partially by official support plus the government using its deposits at the central bank. In the medium and long term, declining grants will be replaced by external concessional loans with 35 percent grant element.
Realism tools flagged	
Mechanical risk rating under the external DSA	Moderate
Mechanical risk rating under the public DSA	Moderate

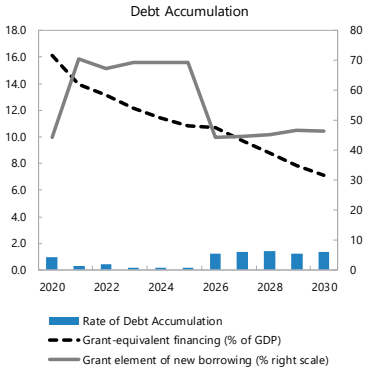
[1] Debt coverage is the same as in the previous DSA which includes central government and central bank borrowing on behalf of the government.
[2] Afghanistan's Composite Indicator score is 2.61 and its debt-carrying capacity is assessed to be weak based on the October 2019 WEO and CPIA 2019.
[3] Metrics used for the long term are averages over the 2025-39 period for the previous DSA and the 2026–40 period for this DSA.

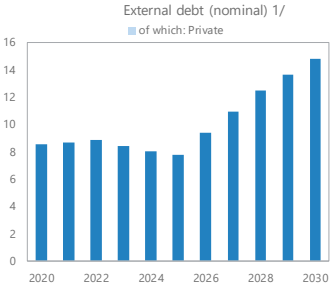
Table 1. Islamic Republic of Afghanistan: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**6.4**	**6.3**	**6.3**	**8.5**	**8.6**	**8.8**	**8.4**	**8.0**	**7.8**	**14.8**	**14.6**	**6.7**	**10.1**
of which: public and publicly guaranteed (PPG)	**6.4**	**6.3**	**6.3**	**8.5**	**8.6**	**8.8**	**8.4**	**8.0**	**7.8**	**14.8**	**14.6**	**6.7**	**10.1**
Change in external debt	0.3	-0.1	0.0	2.2	0.1	0.2	-0.5	-0.4	-0.2	1.2	-0.4		
Identified net debt-creating flows	**-7.5**	**-13.3**	**-9.0**	**-4.8**	**-6.7**	**-5.0**	**-4.1**	**-3.3**	**-2.9**	**6.9**	**7.6**	**-12.5**	**-0.8**
Non-interest current account deficit	**-7.1**	**-13.0**	**-8.6**	**-5.0**	**-5.9**	**-4.2**	**-3.3**	**-2.6**	**-2.2**	**9.4**	**14.1**	**-11.5**	**0.3**
Deficit in balance of goods and services	33.7	30.1	32.3	35.8	33.1	30.8	28.5	26.5	24.9	22.9	16.7	**32.3**	**27.3**
Exports	5.7	9.9	7.4	5.5	6.8	6.7	6.5	6.4	6.4	6.4	5.5		
Imports	39.4	40.0	39.7	41.3	40.0	37.5	35.0	32.9	31.4	29.3	22.2		
Net current transfers (negative = inflow)	-39.5	-42.1	-39.6	-39.4	-37.7	-33.8	-30.7	-28.0	-26.2	-12.8	-2.3	**-43.2**	**-25.9**
of which: official	-37.0	-40.1	-37.4	-38.3	-36.0	-32.2	-29.2	-26.5	-24.6	-10.9	0.0		
Other current account flows (negative = net inflow)	-1.3	-1.1	-1.4	-1.4	-1.3	-1.2	-1.1	-1.1	-1.0	-0.7	-0.3	**-0.6**	**-1.0**
Net FDI (negative = inflow)	**-0.2**	**-0.4**	**-0.5**	**0.0**	**-0.5**	**-0.5**	**-0.5**	**-0.5**	**-0.5**	**-2.1**	**-6.1**	**-0.6**	**-0.9**
Endogenous debt dynamics 2/	**-0.2**	**0.1**	**0.2**	**0.2**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.4**	**-0.4**		
Contribution from nominal interest rate	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1		
Contribution from real GDP growth	-0.2	-0.2	-0.2	0.2	-0.4	-0.4	-0.3	-0.3	-0.3	-0.5	-0.6		
Contribution from price and exchange rate changes	-0.1	0.2	0.3	...	...	...	...	...	...	...	...		
Residual 3/	**7.8**	**13.2**	**9.0**	**7.0**	**6.8**	**5.2**	**3.6**	**3.0**	**2.7**	**-5.7**	**-8.0**	**11.5**	**1.6**
of which: exceptional financing	0.0	0.0	0.0	-1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
CCRT Debt Relief	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**3.8**	**4.9**	**4.9**	**4.8**	**4.5**	**4.3**	**4.1**	**8.3**	**8.6**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**51.2**	**88.6**	**71.9**	**71.8**	**69.4**	**66.3**	**63.5**	**130.1**	**156.9**		
PPG debt service-to-exports ratio	**4.4**	**2.2**	**2.6**	**4.2**	**3.2**	**2.9**	**3.5**	**3.7**	**3.6**	**5.8**	**9.4**		
PPG debt service-to-revenue ratio	**2.1**	**1.7**	**1.4**	**2.3**	**1.7**	**1.4**	**1.5**	**1.5**	**1.4**	**2.1**	**2.8**		
Gross external financing need (Million of U.S. dollars)	-1425.0	-2630.7	-1745.1	-899.1	-1216.0	-981.1	-864.2	-740.6	-691.6	2988.7	7451.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	2.9	2.7	3.0	-3.0	4.5	4.5	4.0	4.0	4.0	4.0	4.0	**4.9**	**3.4**
GDP deflator in US dollar terms (change in percent)	1.0	-3.8	-4.8	-0.6	0.8	5.2	6.1	5.7	3.5	3.1	1.2	**0.0**	**3.0**
Effective interest rate (percent) 4/	0.4	0.4	0.5	0.6	0.5	0.5	0.5	0.5	0.5	1.0	1.0	**0.3**	**0.7**
Growth of exports of G&S (US dollar terms, in percent)	1.4	70.7	-27.0	-27.9	30.2	8.3	7.6	8.2	7.6	6.3	5.7	**1.5**	**6.0**
Growth of imports of G&S (US dollar terms, in percent)	8.6	0.3	-2.7	0.2	1.9	3.2	3.0	3.3	2.5	5.1	5.6	**8.0**	**3.6**
Grant element of new public sector borrowing (in percent)	...	...	...	44.4	70.5	67.1	69.3	69.3	69.3	46.5	65.9	**...**	**56.1**
Government revenues (excluding grants, in percent of GDP)	12.2	13.1	13.6	10.2	12.6	14.0	15.5	15.6	15.9	17.3	18.7	**11.0**	**15.4**
Aid flows (in Million of US dollars) 5/	2670.0	2997.8	2483.3	2976.9	2758.0	2823.5	2905.2	3005.5	3069.7	3101.5	232.7		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	16.1	13.9	13.2	12.1	11.4	10.8	7.1	0.4	**...**	**11.1**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	94.1	98.2	97.5	98.5	98.5	98.6	85.0	65.9	**...**	**92.7**
Nominal GDP (Million of US dollars)	20,187	19,943	19,560	18,853	19,856	21,817	24,072	26,470	28,488	38,931	87,176		
Nominal dollar GDP growth	3.9	-1.2	-1.9	-3.6	5.3	9.9	10.3	10.0	7.6	7.2	5.2	**5.0**	**6.5**
Memorandum items:													
PV of external debt 7/	...	...	3.8	4.9	4.9	4.8	4.5	4.3	4.1	8.3	8.6		
In percent of exports	...	...	51.2	88.6	71.9	71.8	69.4	66.3	63.5	130.1	156.9		
Total external debt service-to-exports ratio	4.4	2.2	2.6	4.2	3.2	2.9	3.5	3.7	3.6	5.8	9.4		
PV of PPG external debt (in Million of US dollars)			738.2	920.7	973.6	1052.2	1093.4	1130.4	1164.8	3244.0	7528.0		
(PVt-PVt-1)/GDPt-1 (in percent)				0.9	0.3	0.4	0.2	0.2	0.1	1.3	0.1		
Non-interest current account deficit that stabilizes debt ratio	-7.4	-12.9	-8.6	-7.2	-6.0	-4.4	-2.9	-2.2	-1.9	8.2	14.5		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

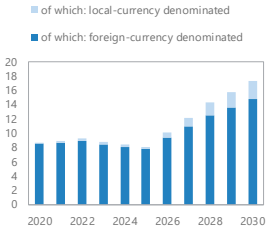
Table 2. Islamic Republic of Afghanistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**7.5**	**6.8**	**6.3**	**8.7**	**8.9**	**9.2**	**8.8**	**8.4**	**8.1**	**17.3**	**20.1**	**7.5**	**11.1**
of which: external debt	**6.4**	**6.3**	**6.3**	**8.5**	**8.6**	**8.8**	**8.4**	**8.0**	**7.8**	**14.8**	**14.6**	**6.7**	**10.1**
Change in public sector debt	-0.4	-0.7	-0.5	2.3	0.3	0.3	-0.4	-0.4	-0.3	1.5	-0.2		
Identified debt-creating flows	**0.0**	**-1.3**	**0.9**	**4.0**	**0.8**	**0.3**	**-0.5**	**-0.4**	**-0.3**	**0.7**	**1.8**	**-0.9**	**0.5**
Primary deficit	**0.6**	**-1.6**	**0.9**	**3.4**	**1.3**	**0.9**	**0.3**	**0.3**	**0.3**	**1.6**	**2.5**	**0.3**	**1.0**
Revenue and grants	25.3	28.1	26.1	25.5	25.9	26.5	27.3	26.7	26.4	23.4	18.7	**24.6**	**25.7**
of which: grants	13.1	15.0	12.5	15.3	13.3	12.5	11.7	11.0	10.5	6.0	0.0		
Primary (noninterest) expenditure	25.9	26.5	27.0	28.9	27.3	27.4	27.6	27.0	26.7	24.9	21.2	**24.9**	**26.7**
Automatic debt dynamics	**-0.4**	**0.3**	**0.0**	**0.6**	**-0.6**	**-0.7**	**-0.8**	**-0.7**	**-0.5**	**-0.8**	**-0.7**		
Contribution from interest rate/growth differential	-0.4	-0.3	-0.3	0.1	-0.5	-0.5	-0.5	-0.5	-0.4	-0.7	-0.8		
of which: contribution from average real interest rate	-0.2	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.0		
of which: contribution from real GDP growth	-0.2	-0.2	-0.2	0.2	-0.4	-0.4	-0.4	-0.3	-0.3	-0.6	-0.8		
Contribution from real exchange rate depreciation	0.0	0.5	0.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.2**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.7**	**0.0**
Privatization receipts (negative)	-0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.4**	**0.6**	**-1.4**	**-1.2**	**-0.6**	**-0.1**	**-0.3**	**-0.2**	**-0.2**	**0.6**	**-1.9**	**-0.1**	**0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**3.8**	**5.3**	**5.3**	**5.4**	**5.1**	**4.9**	**4.6**	**10.8**	**14.2**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**14.6**	**20.7**	**20.6**	**20.5**	**18.8**	**18.2**	**17.3**	**46.4**	**76.3**		
Debt service-to-revenue and grants ratio 3/	**3.2**	**2.9**	**2.5**	**1.0**	**0.9**	**0.8**	**1.3**	**1.5**	**1.3**	**4.6**	**11.8**		
Gross financing need 4/	1.4	-0.8	1.6	3.6	1.6	1.1	0.7	0.7	0.6	2.6	4.8		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	2.9	2.7	3.0	-3.0	4.5	4.5	4.0	4.0	4.0	4.0	4.0	**4.9**	**3.4**
Average nominal interest rate on external debt (in percent)	0.4	0.4	0.6	0.6	0.6	0.5	0.5	0.5	0.5	1.0	1.0	**0.3**	**0.7**
Average real interest rate on domestic debt (in percent)	-5.0	0.8	-0.2	-4.6	2.4	1.4	0.9	0.4	2.1	2.1	2.1	**-3.0**	**1.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	0.8	8.7	4.2	...	...	...	...	...	...	...	...	**2.7**	**...**
Inflation rate (GDP deflator, in percent)	5.2	1.2	2.4	4.8	4.5	5.6	6.0	6.5	4.8	4.8	4.8	**4.1**	**5.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.2	6.6	4.8	3.7	-1.4	5.0	4.6	1.7	3.0	2.3	4.0	**8.3**	**2.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.9	-0.9	1.4	1.1	1.1	0.6	0.7	0.7	0.6	0.0	2.7	**0.5**	**0.0**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Islamic Republic of Afghanistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020-30

(In percent)

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	4.9	4.9	4.8	4.5	4.3	4.1	5.1	6.0	6.9	7.6	8.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	4.9	1.3	-3.2	-8.4	-14.1	-20.1	-26.6	-34.4	-43.5	-53.7	-64.7
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B. Bound Tests											
B1. Real GDP growth	4.9	5.2	5.5	5.1	4.8	4.6	5.8	6.8	7.8	8.6	9.4
B2. Primary balance	4.9	5.3	5.6	5.3	5.1	5.0	6.1	7.0	7.9	8.6	9.3
B3. Exports	4.9	7.2	10.0	9.5	8.9	8.6	9.7	10.5	11.3	11.9	12.5
B4. Other flows 2/	4.9	8.6	12.0	11.4	10.7	10.3	11.5	12.2	12.9	13.4	14.0
B6. One-time 30 percent nominal depreciation	4.9	6.2	-0.6	-0.6	-0.6	-0.7	0.6	1.9	3.2	4.2	5.3
B6. Combination of B1-B5	4.9	10.4	11.6	10.9	10.3	9.9	11.1	12.0	12.8	13.4	14.1
C. Tailored Tests											
C1. Combined contingent liabilities	4.9	7.8	7.6	7.2	7.6	7.3	8.5	9.5	10.3	10.9	11.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	88.6	71.9	71.8	69.4	66.3	63.5	77.9	92.0	106.5	117.7	130.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	88.6	19.2	-47.7	-128.6	-219.6	-311.9	-403.7	-523.8	-667.3	-831.2	-1010.6
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B. Bound Tests											
B1. Real GDP growth	88.6	71.9	71.8	69.4	66.3	63.5	77.9	92.0	106.5	117.7	130.1
B2. Primary balance	88.6	78.0	83.8	81.2	79.7	78.3	92.7	106.9	121.7	132.7	**144.9**
B3. Exports	88.6	**205.7**	**480.8**	**464.9**	**446.0**	**429.6**	**473.9**	**514.6**	**558.1**	**591.2**	**627.9**
B4. Other flows 2/	88.6	126.3	**179.3**	**173.4**	**166.5**	**160.6**	**173.6**	**185.2**	**197.8**	**207.3**	**218.0**
B6. One-time 30 percent nominal depreciation	88.6	71.9	-7.4	-7.3	-7.5	-8.0	7.5	23.3	39.2	51.7	65.4
B6. Combination of B1-B5	88.6	**222.5**	**156.5**	**306.2**	**293.9**	**283.2**	**310.1**	**334.7**	**361.1**	**381.0**	**403.3**
C. Tailored Tests											
C1. Combined contingent liabilities	88.6	113.8	112.9	110.0	117.9	114.0	128.1	**144.4**	**158.1**	**168.6**	**181.0**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	4.2	3.2	2.9	3.5	3.7	3.6	6.0	5.9	5.9	5.9	5.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	4.2	3.2	1.8	1.1	-0.4	-2.2	-1.2	-3.6	-6.4	-9.7	-13.3
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B. Bound Tests											
B1. Real GDP growth	4.2	3.2	2.9	3.5	3.7	3.6	6.0	5.9	5.9	5.9	5.8
B2. Primary balance	4.2	3.2	3.0	3.8	3.9	3.9	6.3	6.2	6.2	6.2	6.1
B3. Exports	4.2	6.4	11.8	16.9	17.1	16.5	24.2	23.8	23.6	23.2	22.8
B4. Other flows 2/	4.2	3.2	4.1	5.9	5.9	5.7	8.0	7.8	7.7	7.6	7.4
B6. One-time 30 percent nominal depreciation	4.2	3.2	2.9	1.8	2.0	2.0	4.5	4.5	4.6	4.6	4.6
B6. Combination of B1-B5	4.2	5.0	9.0	10.8	10.9	10.5	15.2	15.0	14.8	14.6	14.3
C. Tailored Tests											
C1. Combined contingent liabilities	4.2	3.2	3.8	4.4	4.5	4.7	7.0	6.9	6.9	6.9	6.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	2.3	1.7	1.4	1.5	1.5	1.4	2.4	2.3	2.2	2.2	2.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	2.3	1.8	0.9	0.5	-0.1	-0.9	-0.5	-1.4	-2.4	-3.6	-4.9
	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B. Bound Tests											
B1. Real GDP growth	2.3	1.9	1.6	1.7	1.7	1.6	2.7	2.6	2.5	2.5	2.4
B2. Primary balance	2.3	1.7	1.5	1.6	1.6	1.6	2.5	2.4	2.4	2.3	2.2
B3. Exports	2.3	1.8	1.8	2.2	2.2	2.1	3.0	2.9	2.8	2.7	2.6
B4. Other flows 2/	2.3	1.7	2.0	2.5	2.4	2.3	3.1	3.0	2.9	2.8	2.7
B6. One-time 30 percent nominal depreciation	2.3	2.2	1.8	0.9	1.1	1.0	2.2	2.2	2.2	2.2	2.1
B6. Combination of B1-B5	2.3	1.9	2.3	2.5	2.4	2.3	3.3	3.1	3.1	3.0	2.9
C. Tailored Tests											
C1. Combined contingent liabilities	2.3	1.7	1.8	1.9	1.9	1.9	2.8	2.7	2.6	2.6	2.5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

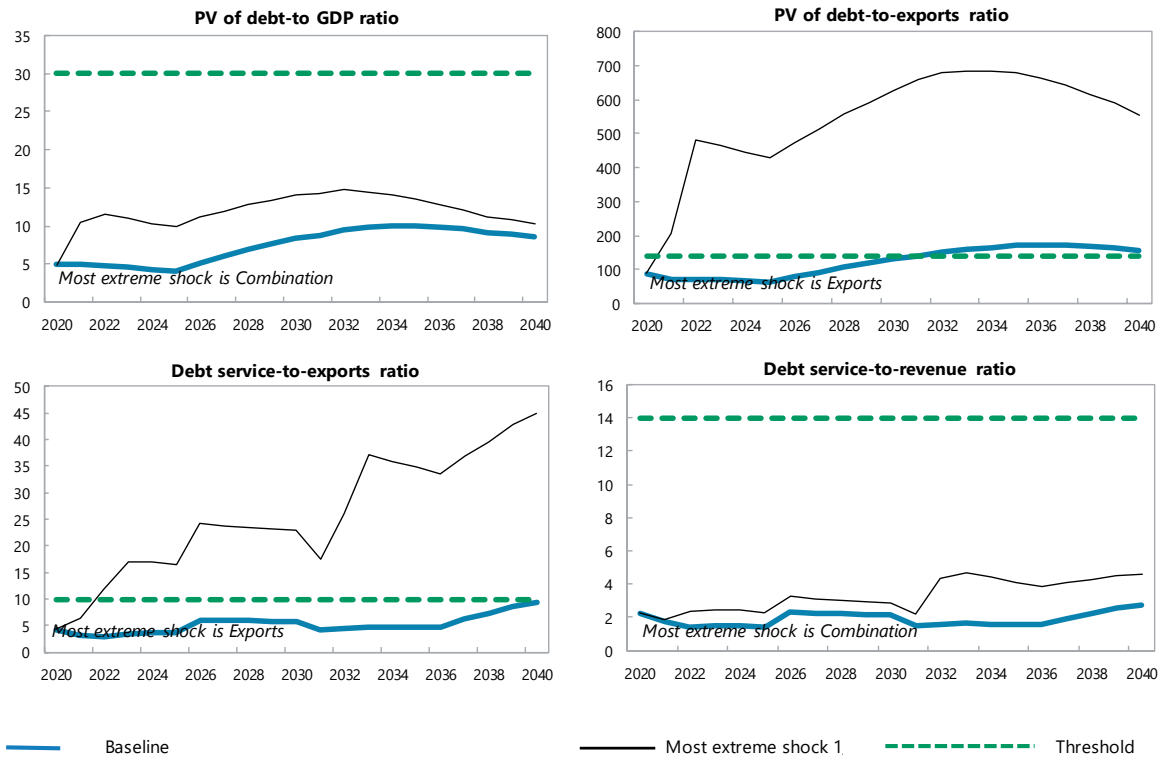
1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Islamic Republic of Afghanistan: Sensitivity Analysis for Key Indicators of Public Debt, 2020-30

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	5.3	5.3	5.4	5.1	4.9	4.6	5.9	7.3	8.8	9.8	10.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	5	5	4	4	4	4	5	6	7	7	7
B. Bound Tests											
B1. Real GDP growth	5	6	8	9	10	11	14	17	20	22	25
B2. Primary balance	5	6	7	7	6	6	7	8	10	11	12
B3. Exports	5	8	11	10	9	9	10	12	13	14	15
B4. Other flows 2/	5	9	13	12	12	11	12	13	15	16	16
B6. One-time 30 percent nominal depreciation	5	6	5	4	3	2	2	3	3	3	3
B6. Combination of B1-B5	5	6	6	5	5	4	6	7	9	10	11
C. Tailored Tests											
C1. Combined contingent liabilities	5	11	11	10	9	9	10	11	12	13	14
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	20.7	20.6	20.5	18.8	18.2	17.3	22.3	28.5	35.3	40.6	46.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	20.7	17.5	15.8	14.4	14.2	13.5	18.2	23.3	28.0	30.2	31.9
B. Bound Tests											
B1. Real GDP growth	20.7	23.5	28.8	31.9	36.3	40.2	50.7	63.3	76.8	89.1	101.9
B2. Primary balance	20.7	23.5	26.1	24.0	23.0	21.8	26.6	32.8	39.5	44.9	50.6
B3. Exports	20.7	29.1	40.1	36.6	35.6	34.3	38.8	45.1	51.9	57.3	63.0
B4. Other flows 2/	20.7	35.3	48.9	44.6	43.4	41.9	46.2	52.5	59.3	64.7	70.5
B6. One-time 30 percent nominal depreciation	20.7	24.5	20.0	15.4	11.9	8.7	9.1	10.4	12.1	12.5	13.1
B6. Combination of B1-B5	20.7	22.5	22.3	18.4	17.8	16.9	21.9	28.1	34.8	40.1	45.8
C. Tailored Tests											
C1. Combined contingent liabilities	20.7	41.0	39.7	36.5	33.9	32.6	37.2	42.9	49.7	55.1	60.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	1.0	0.9	0.8	1.3	1.5	1.3	2.0	2.2	2.0	3.8	4.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	1.0	0.9	0.7	1.1	0.2	0.5	1.8	1.7	1.6	3.4	3.6
B. Bound Tests											
B1. Real GDP growth	1.0	0.9	1.0	1.7	2.9	3.9	5.1	5.8	6.2	8.7	10.1
B2. Primary balance	1.0	0.9	0.9	1.6	2.7	2.6	2.2	2.7	2.4	4.0	4.8
B3. Exports	1.0	0.9	1.0	1.7	1.8	1.7	2.4	2.5	2.3	4.1	4.9
B4. Other flows 2/	1.0	0.9	1.1	1.9	2.0	1.8	2.5	2.7	2.4	4.3	5.0
B6. One-time 30 percent nominal depreciation	1.0	1.0	1.0	1.4	1.5	0.2	2.1	2.2	1.9	3.5	4.2
B6. Combination of B1-B5	1.0	0.9	0.8	1.3	1.5	1.3	2.0	2.2	2.0	3.8	4.6
C. Tailored Tests											
C1. Combined contingent liabilities	1.0	0.9	1.7	2.1	9.2	2.0	2.7	4.8	2.5	4.4	5.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

Figure 1. Islamic Republic of Afghanistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2020-40



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	0.9%	1.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	37	20
Avg. grace period	9	10

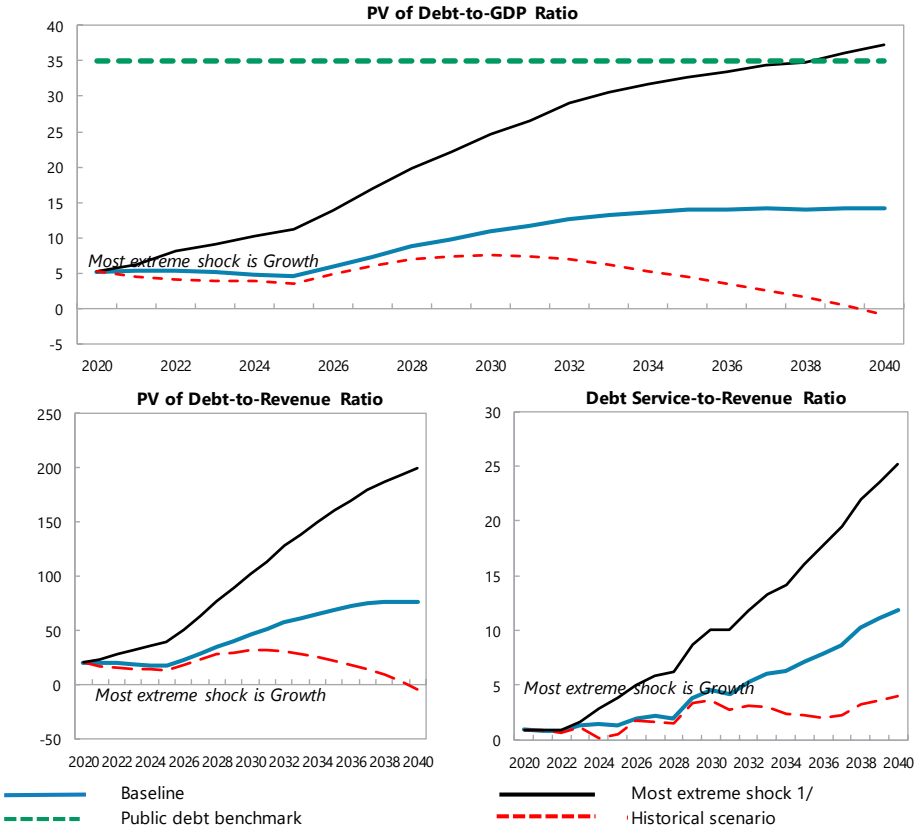
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Islamic Republic of Afghanistan: Indicators of Public Debt Under Alternatives Scenarios, 2020-40



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	65%	65%
Domestic medium and long-term	35%	35%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	0.9%	1.5%
Avg. maturity (incl. grace period)	37	20
Avg. grace period	9	10
Domestic MLT debt		
Avg. real interest rate on new borrowing	1.9%	1.9%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	0%	0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

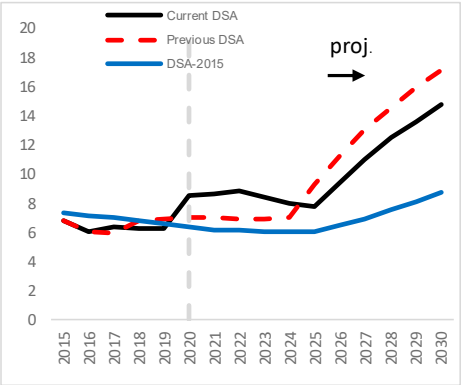
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

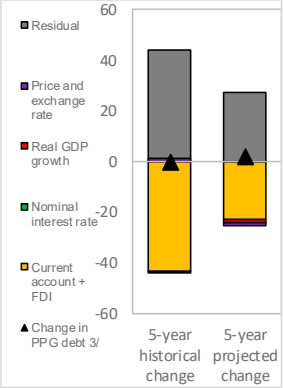
Figure 3. Islamic Republic of Afghanistan: Drivers of Debt Dynamics—Baseline Scenario

External debt

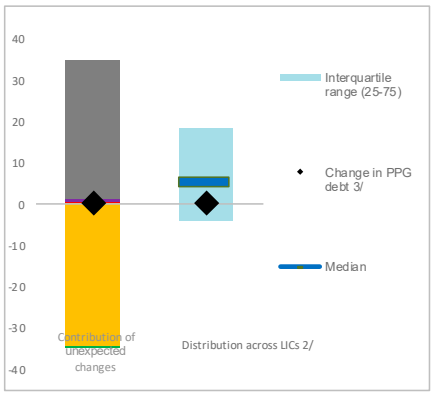
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

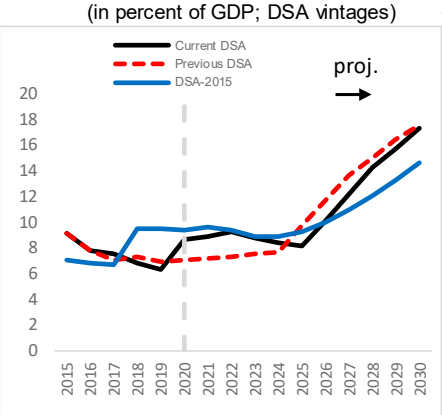


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

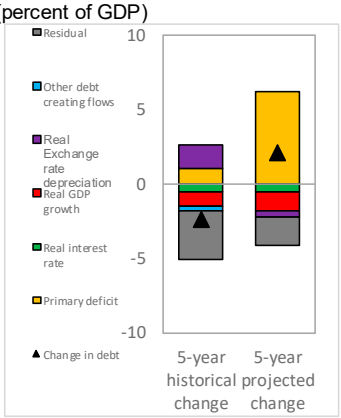


Public debt

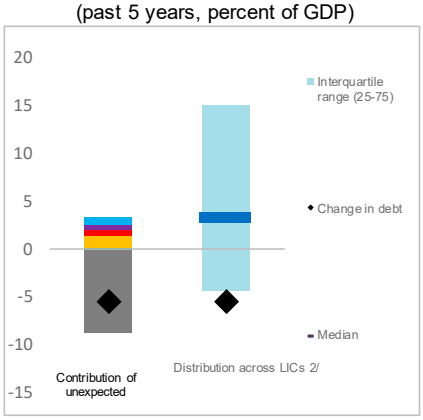
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



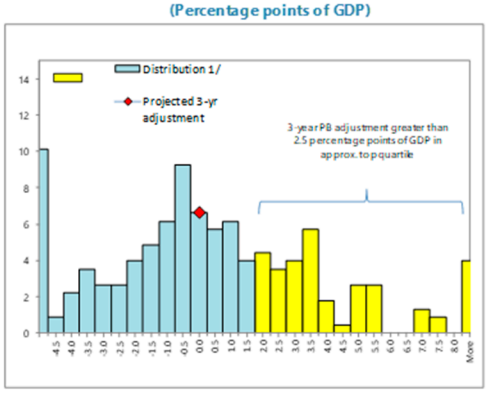
Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

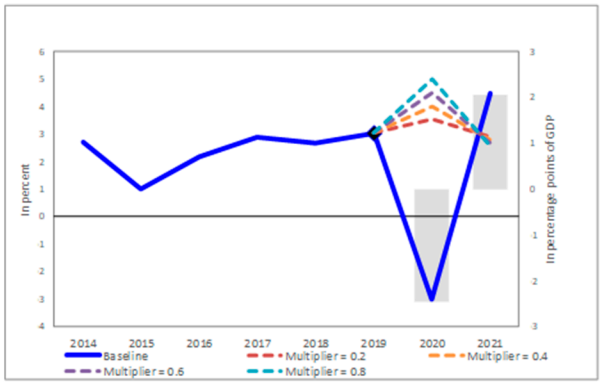
Figure 4. Islamic Republic of Afghanistan: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

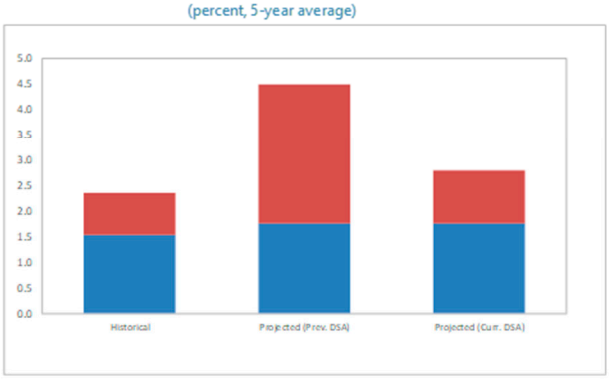
Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates 1/
(% of GDP)

Contribution to Real GDP growth
(percent, 5-year average)



Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA

Contribution of other factors
Contribution of government capital

1/ Historical private and public investment rates are different from previous DSA due to data revisions.